

DataPulse v0 — Warehouse Runbook

Purpose

DataPulse v0 validates that customer behavioral events arriving in BigQuery are complete, trustworthy, identity-aware, and usable for analytics.

Source: Google GA4 public ecommerce dataset

Analysis window: 2021-01-18 to 2021-01-31

Journey modeled: `page_view` → `view_item` → `add_to_cart` → `begin_checkout` → `purchase`

Because the public GA4 dataset contains no populated `user_id`, a deterministic simulated identity layer is used for the CDP identity-stitching demonstration. First `begin_checkout` is treated as the identification point.

Warehouse Structure

RAW — `datapulse_raw`

- `raw_ga4_events` — 264,485 source-like events preserved for traceability.

STAGING — `datapulse_staging`

- `identity_map` — 1,133 anonymous-to-known simulated identity mappings.
- `stg_events` — standardized events with `anonymous_id`, `user_id_at_event`, `stitched_user_id`, identity timestamp, device, geography, and transaction attributes.

MART — `datapulse_mart`

- `events` — business-ready standardized events with purchase-quality flags.
- `users` — one row per visitor with identity status and first/last seen.
- `customer_funnel` / `funnel_summary` — sequential conversion model.
- `daily_event_volume` — dashboard-ready daily event metrics.

Validation & Data Health

Source reconciliation

- Source: 264,485 events
- Raw warehouse: 264,485 events
- Variance: 0

Critical fields

- 0 missing event dates
- 0 missing timestamps
- 0 missing event names
- 0 missing anonymous identities

Duplicates

- 264,485 total rows
- 264,485 distinct full-row hashes
- 0 exact duplicates

Schema

- 27 distinct properties
- 0 casing, whitespace, or naming-format violations
- Event/property naming conforms to `lowercase_snake_case`

Purchase quality

- 738 purchase-labelled events
- 468 valid transactions
- 270 incomplete purchase signals with `(not set)` transaction IDs and missing revenue/item quantity

Incomplete events remain preserved for traceability but are excluded from trusted purchase/revenue metrics.

Identity

- 1,133 mappings
- 1,133 unique anonymous IDs
- 1,133 unique generated user IDs
- 0 mapping collisions

CDP Proof of Value

Sequential customer funnel:

Stage	Users	Previous-stage conversion
Page View	42,885	100.00%
View Item	10,994	25.64%
Add to Cart	1,956	17.79%
Begin Checkout	859	43.92%
Valid Purchase	360	41.91%

Overall page-view-to-valid-purchase conversion: 0.84%.

Operational Monitoring

Daily

- Event arrival and counts by type
- Expected-date continuity
- Critical-field completeness
- Duplicate detection
- Unexpected event/property names
- Purchase transaction/revenue completeness
- Material event-volume anomalies

Weekly

- Tracking-plan/schema changes
- Identity mapping integrity
- Funnel conversion and drop-off trends
- Recurring source-quality issues
- Data dictionary and validation-rule updates

If validation fails, preserve the raw event, identify whether the issue originates in the source or transformation layer, document downstream impact, and prevent unreliable records from entering trusted metrics until resolved.

Known Limitation

Known-user identity is simulated because the public GA4 dataset does not expose populated `user_id`. In production, identity resolution would use actual Segment authentication/`identify` events and schema validation would be enforced against an approved tracking plan.